

	to gravitational potential energy, and its kinetic energy decreases from a maximum to 0 J, while the elastic potential energy decreases to a minimum.
3(a)	Weight is the force acting on the mass due to the gravitational force of attraction by the Earth, pointing downward and directly to the centre of the Earth. The magnitude is mg , where g is the gravitational field strength.

3(b)	The net force acting on the picture is zero. By Newton's first law, the picture will remain at rest if it is originally at rest.
3(c)	Upward force = downward force $2T \sin 45^\circ = 42$ $T = \frac{42}{2 \sin 25^\circ}$ $= 49.7 \text{ N}$ $\approx 50 \text{ N (shown)}$
3(d)(i)	B: length = 45 cm $\theta = \cos^{-1} \left(\frac{60}{72.5} \right) = 34.14^\circ$ $2T_B \sin \theta = 42$ $T_B = 37.4 \text{ N} < 40 \text{ N}$ C: length = 130 cm $\theta = \cos^{-1} \left(\frac{60}{65} \right) = 22.62^\circ$ $2T_C \sin \theta = 42$ $T_C = 54.6 \text{ N} > 50 \text{ N}$ D: length = 125 cm $\theta = \cos^{-1} \left(\frac{60}{62.5} \right) = 16.26^\circ$ $2T_D \sin \theta = 42$ $T_D = 75 \text{ N} > 60 \text{ N}$ Only wire B is able to withstand the breaking tension, thus only wire B is suitable.
3(d)(ii)	Even if wire A is very long so that the wire makes an angle of 90° with the horizontal, the tension of $A = 42/2 = 21 \text{ N}$, which is still larger than 20 N , the breaking tension. So it is not possible for longer wire A to support the weight.
3(e)(i)	Breaking stress $= \frac{60}{\frac{\pi (1.56 \times 10^{-3})^2}{4}}$ $= 3.14 \times 10^7 \text{ N m}^{-2}$
3(e)(ii)	A and D as they have the same breaking stress.
3(f)(i)	Percentage uncertainty $= \frac{0.01}{1.56} \times 100 = 0.64 \%$
3(f)(ii)	$A = \frac{\pi d^2}{4} = 1.91 \times 10^{-6} \text{ m}^2$

	$\frac{\Delta A}{A} = \frac{2\Delta d}{d} = \frac{2(0.64)}{100} = 0.0128$ $\Delta A = 0.0128A$ $= 0.0128(1.91 \times 10^{-6})$ $= 2 \times 10^{-8} \text{ m}^2$
4(c)	From defining equation of Magnetic Flux Density